Aditya Manivel

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EDUCATION

University of Washington ***September 2024 – June 2028***
Bachelor of Science, Electrical & Computer Engineering *Seattle, WA*

Coursework: Circuit Theory, Electronic Devices, Digital Circuits, Computer Hardware Skills, Fundamentals of Electrical Engineering, Python for Signal Processing, Object-Oriented Programming, Data Structures & Algorithms.

SKILLS

- **Embedded & EDA:** STM32, Arduino, ESP32, Altium Designer, KiCad
- **Programming Languages & OS:** C, Python, Java, Linux, Bare Metal, PowerShell
- **Protocols:** UART, I2C, SPI, HTTP, IP, TCP
- **Laboratory Skills:** Bench Power Supply, Oscilloscope, Logic Analyzer, Multimeter, Soldering, Crimping

EXPERIENCE

Husky Robotics Club ***September 2024 – Present***
Hardware Engineer *Seattle, WA*

- Collaborated with the electronics team to design multilayer PCBs in Altium Designer for the team's mockup Mars Rover.
- Currently designing a custom FPGA-based PCB, targeting an 8-layer stackup and integrating an Efinix Titanium Ti375 FPGA.
- Designed a 20 A, 24 V three-phase MOSFET bridge with high-/low-side gate drivers and PWM control for a brushless DC motor board as a custom alternative to commercial motor controllers for the rover, projected to reduce system cost by 30%.
- Participated in PCB design reviews and supported board bring-up. Assembled PCBs using through-hole and reflow soldering.

Spacelabs Healthcare ***June 2026 – September 2026***
Hardware Test Engineering Intern *Snoqualmie, WA*

- Redesigned the electronics architecture for 5 automated test fixtures used to evaluate medical-device touchscreen reliability across their product life cycle. Presented results to company leadership.
- Designed a 4-layer STM32-based PCB in Altium Designer supporting 3 motors, 2 buck regulators, and 4 limit switches.
- Created a 2-board system consisting of a PCB and Linux-based development board. Defined system-level power distribution and communication between the boards.
- Developed STM32 firmware and Linux-based software using Claude-assisted development for motor control, limit-switch monitoring, UART communication, web interface for automated test execution, remote access, and system monitoring.
- Coordinated PCB fabrication and assembly with external vendors, performed board bring-up and hardware testing, and integrated firmware to validate system functionality.

Programmable Matter Lab ***January 2026 – July 2026***
Undergraduate Researcher *Seattle, WA*

- Worked in the Programmable Matter Lab to develop MatterMirror, a tile based capacitive sensing system that enables bidirectional mapping between physical block configurations and software models.
- Prototyped RC-based circuits on breadboards. Implemented Bluetooth communication between tiles and host computer.
- Designed initial 2-layer tile PCB in KiCad integrating a Seeed Studio XIAO ESP32-C3 board, 16-channel analog multiplexer, and capacitance-measurement circuitry.

Human Powered Submarine Team ***September 2024 – June 2025***
Electrical Engineer *Seattle, WA*

- Integrated a Qi-standard inductive wireless charging system into the sealed electronics module of a pedal-driven submarine, enabling wireless battery charging without disassembling the waterproof enclosure.
- Completed electrical integration and system wiring of the wireless charger, battery management system (BMS), and a 12,000 mAh battery pack (4 × 18650 Li-ion cells, 3.7 V nominal), supporting an estimated 44 hours of continuous operation.
- Implemented a waterproof electronics system using sealed enclosures and cable penetrators, with external electronics encapsulated in epoxy.

PROJECTS

AI Classifier ***September 2026***

- Developed a real-time hand-gesture classification system using an AI-capable Arduino UNO Q, USB camera, Python, OpenCV, and a MobileNetV2-based transfer-learning model.
- Used an online dataset, achieving 86.56% test-set accuracy with the int8 implementation. Deployed model using Edge Impulse.

Line-Following Robot ***November 2025***

- Designed and fabricated a custom light-sensing PCB in KiCad, integrating LEDs, photoresistors, and supporting circuitry for an Arduino-based autonomous line-following robot.
- Assembled the PCB using through-hole soldering, integrated sensors with motor-control electronics, and tuned PID control parameters to improve autonomous line tracking.